

APPLIED PHYSICS

Week 3 Study Notes

Oscillations & Simple Harmonic Motion

Instructor

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Course

Applied Physics

Week

Week 3 — Lecture A & B

Topics Covered

Sections 15-2 · 15-3 · 15-4 + Practice Problems & Formula Reference

Contents

Section 15-2 | Simple Harmonic Motion — displacement, velocity, acceleration, phasor

Section 15-3 | The Force Law — Hooke's Law, ODE derivation, isochronism

Section 15-4 | Energy in SHM — PE, KE, total energy, v-x semi-ellipse

Sample Problems | Four fully worked examples

Practice Problems | Four problems with complete solutions

Formula Reference | Complete tables: kinematics, force, energy

Simple Harmonic Motion

Displacement, velocity, acceleration, frequency, period, phase, and phasor interpretation

Any motion that repeats at regular intervals is called **periodic motion** or **harmonic motion**. Simple Harmonic Motion (SHM) is the special case where displacement varies sinusoidally with time. It is the most fundamental oscillatory motion in physics and appears in springs, molecular vibrations, AC circuits, and sound waves.

15-2 A | SHM as Projection of Circular Motion

One of the most elegant ways to understand SHM is as the **x-projection of a point rotating at constant angular speed ω in a circle** of radius A . As the point rotates, its horizontal shadow moves back and forth — that is exactly SHM.

WHY THE PHASOR CIRCLE EXPLAINS EVERYTHING

Radius = Amplitude A : The shadow's maximum displacement equals the circle radius. This is why A is always positive.

ω = rotation rate: How fast the point rotates equals the angular frequency of the SHM.

ϕ = starting angle: The angle of the point on the circle at $t = 0$ is the phase constant ϕ . It sets the initial position and direction.

Cosine and sine appear naturally: The x-projection is cosine ($\rightarrow$ displacement); the y-projection is sine ($\rightarrow$ velocity). Both are SHM, 90° apart.

15-2 B | Basic Terminology & Definitions

Quantity	Symbol	Meaning	Unit
Equilibrium position	$x = 0$	Where net force = 0; natural rest position	m
Displacement	x	Signed distance from equilibrium	m
Amplitude	A (or x_m)	Maximum displacement — always positive	m
Period	T	Time for one complete oscillation (cycle)	s
Frequency	f	Oscillations completed per second = $1/T$	Hz
Angular frequency	ω	Radians per second = $2\pi f$	rad/s
Phase constant	ϕ	Sets initial position/direction at $t = 0$	rad

Frequency f (Eq. 15-1)

$$1 \text{ hertz} = 1 \text{ Hz} = 1 \text{ oscillation/second} = 1 \text{ s}^{-1}$$

f = number of complete oscillations per second. SI unit: hertz (Hz).

Period T (Eq. 15-2)

$$T = 1/f$$

 T = time for one full oscillation. T and f are reciprocals.

15-2 C | Displacement of SHM

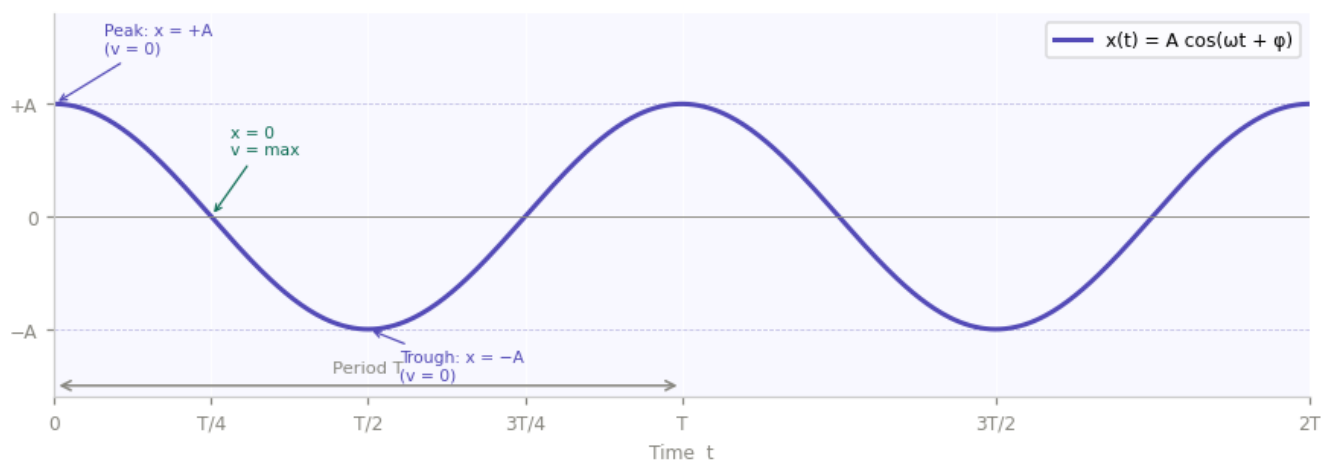
The position of a particle in SHM as a function of time is a cosine function:

Displacement (Eq. 15-3)

$$x(t) = x_m \cos(\omega t + \phi)$$

 x_m = amplitude; ω = angular frequency (rad/s); ϕ = phase constant; $(\omega t + \phi)$ = phase of the motion at time t Every feature of the cosine curve has physical meaning (for $\phi = 0$):

Feature of $x(t)$	Value	When it occurs	Physical meaning
Crest (peak)	$x = +x_m$	$t = 0, T, 2T \dots$	Furthest right — mass momentarily stopped
Trough	$x = -x_m$	$t = T/2, 3T/2 \dots$	Furthest left — mass momentarily stopped
Zero crossing	$x = 0$	$t = T/4, 3T/4 \dots$	Back at equilibrium — maximum speed
One full period T	x repeats	$t = 0$ to T	Motion is identical in every period



Graph 1 — Displacement $x(t) = A \cos(\omega t + \phi)$

Purple curve: the cosine wave showing how displacement varies with time.

Peaks (+A): block is at rightmost position, momentarily stopped ($v = 0$).

Troughs (-A): block at leftmost position, also stopped.

Zero crossings ($x = 0$): block passes through equilibrium at maximum speed.Double-headed arrow marks one full period T . The pattern then repeats exactly.Try mentally sliding the curve left/right — that is what changing phase ϕ does.

15-2 C(i) | Derivation 1 — From Newton's Law to $x(t)$: The 11-Step ODE

We combine Newton's Second Law $F = ma$ with Hooke's Law $F = -kx$ to obtain the equation of motion, then solve it step by step to prove $x(t) = A \cos(\omega t + \phi)$.

Step	Equation / Result	Explanation
Step 1 Newton's 2nd Law	$F = m \cdot a$	Net force = mass $\times$ acceleration. Always our starting point.
Step 2 Apply Hooke's Law	$F = -k x$	The only force is the spring. k = spring constant (N/m). Negative sign: force opposes displacement.
Step 3 Equate both expressions	$m \cdot a = -k x$	Substitute Hooke's Law for F into Newton's Second Law.
Step 4 Write a as d^2x/dt^2	$m \cdot d^2x/dt^2 = -k x$	Acceleration a is the second derivative of displacement x with respect to time.
Step 5 Rearrange to ODE form	$d^2x/dt^2 + (k/m)x = 0$	Divide both sides by m and move the kx/m term to the left.
Step 6 Define $\omega^2 = k/m$	$d^2x/dt^2 + \omega^2 x = 0$	Let $\omega^2 = k/m$. This is the standard harmonic oscillator differential equation.
Step 7 Propose trial solution	$x(t) = A \cos(\omega t + \phi)$	Physical intuition: oscillating motion is sinusoidal. A and ϕ are fixed by initial conditions.
Step 8 Differentiate once	$dx/dt = -A\omega \sin(\omega t + \phi)$	First derivative using chain rule: $d/dt[\cos(\omega t + \phi)] = -\omega \sin(\omega t + \phi)$.
Step 9 Differentiate again	$d^2x/dt^2 = -A\omega^2 \cos(\omega t + \phi)$	Second derivative. Chain rule again: $d/dt[-\sin(\omega t + \phi)] = -\omega \cos(\omega t + \phi)$.
Step 10 Observe the identity	$d^2x/dt^2 = -\omega^2 \cdot [A \cos(\omega t + \phi)] = -\omega^2 x$	The second derivative equals $-\omega^2 \times$ the original function. This is the defining property of cosine.
Step 11 Substitute into ODE ✓	$(-\omega^2 x) + \omega^2 x = 0 \quad \checkmark$	Left side = 0. The trial solution satisfies the equation. Derivation complete.

WHY IS ACCELERATION THE SECOND DERIVATIVE OF DISPLACEMENT?

Step 1 — Position to Velocity (1st derivative): Velocity is the rate of change of position with respect to time. In calculus notation: $v = dx/dt$. This means “how fast is position x changing per unit time?”

Step 2 — Velocity to Acceleration (2nd derivative): Acceleration is the rate of change of velocity: $a = dv/dt$. Substituting $v = dx/dt$ gives $a = d/dt(dx/dt) = d^2x/dt^2$. The notation d^2x/dt^2 means “differentiate x twice with respect to t .”

The full chain: $x(t) \rightarrow v = dx/dt \rightarrow a = dv/dt = d^2x/dt^2$

Worked example with $x(t) = A \cos(\omega t)$: Taking the 1st derivative gives $v = -A\omega \sin(\omega t)$. Taking the 2nd derivative gives $a = -A\omega^2 \cos(\omega t)$. Notice that differentiating cosine twice brings out a factor of $(-\omega^2)$ and returns the same cosine function — this is exactly why $a = -\omega^2 x$.

Why this matters in Step 4 of the ODE: Newton's 2nd Law $F = ma$ uses acceleration a . Since $a = d^2x/dt^2$, the equation $m \cdot a = -kx$ becomes $m \cdot d^2x/dt^2 = -kx$. This is a differential equation — an equation involving x and its derivatives — and solving it reveals how the block actually moves in time.

Physical intuition: Position tells you WHERE the block is. Velocity (1st derivative) tells you how fast it is moving. Acceleration (2nd derivative) tells you how fast the velocity itself is changing — i.e., how

hard the spring is pushing. All three are linked: $F = -kx$ determines a , which changes v , which changes x .

RESULT — General Solution: Displacement Equation

$$x(t) = A \cos(\omega t + \phi)$$

A = amplitude (m) | ω = angular frequency (rad/s) = $\sqrt{k/m}$
 | ϕ = phase constant (rad)

Verification: differentiate twice $\rightarrow d^2x/dt^2 = -A\omega^2 \cos(\omega t + \phi) = -\omega^2 x(t)$ ✓

This is the complete general solution. Every SHM system has this form.

15-2 D | Angular Frequency — Derivation

Since displacement must repeat every period T , we require $x(t) = x(t+T)$:

DERIVATION — Angular Frequency ω (Eq. 15-5)

Start: $x(t) = x_m \cos(\omega t + \phi)$

After one period T : $x(t+T) = x_m \cos(\omega(t+T) + \phi)$

For these to be equal, cosine argument must increase by 2π :

$$\omega(t+T) + \phi = \omega t + \phi + 2\pi$$

$$\omega T = 2\pi$$

$$\omega = 2\pi/T = 2\pi f \quad \leftarrow \text{Eq. 15-5}$$

Units: rad/s (radians per second)

Angular Frequency ω (Eq. 15-5)

$$\omega = 2\pi/T = 2\pi f$$

ω , f , and T all describe the same oscillation rate: $\omega = 2\pi f = 2\pi/T \Leftrightarrow T = 1/f = 2\pi/\omega$

15-2 D(i) | Why $\omega T = 2\pi$ — The Angular Frequency Derivation Explained

The formula $\omega T = 2\pi$ is not something to memorise — it follows logically from one simple fact: SHM completes one full oscillation in time T , which on the phasor circle means one full revolution, and one full revolution is always exactly 2π radians.

DERIVATION — Why $\omega T = 2\pi$

Step 1 — What is ω physically? Imagine a point spinning around a circle at constant speed. ω is how fast it spins — in radians per second. One full circle = 2π radians. So $\omega = 2\pi$ rad/s means one complete revolution per second.

Step 2 — SHM is the x-shadow of circular motion (phasor). $x(t) = A \cos(\omega t)$ is just the horizontal projection of the spinning point. The circle radius = amplitude A . The angle at time $t = \omega t$ radians.

Step 3 — After one period T , the block has returned to its starting position. On the circle, the spinning point has completed exactly one full revolution. Total angle swept = $\omega \times T$.

Step 4 — One full revolution = 2π radians. For the motion to repeat, the swept angle must equal exactly 2π :

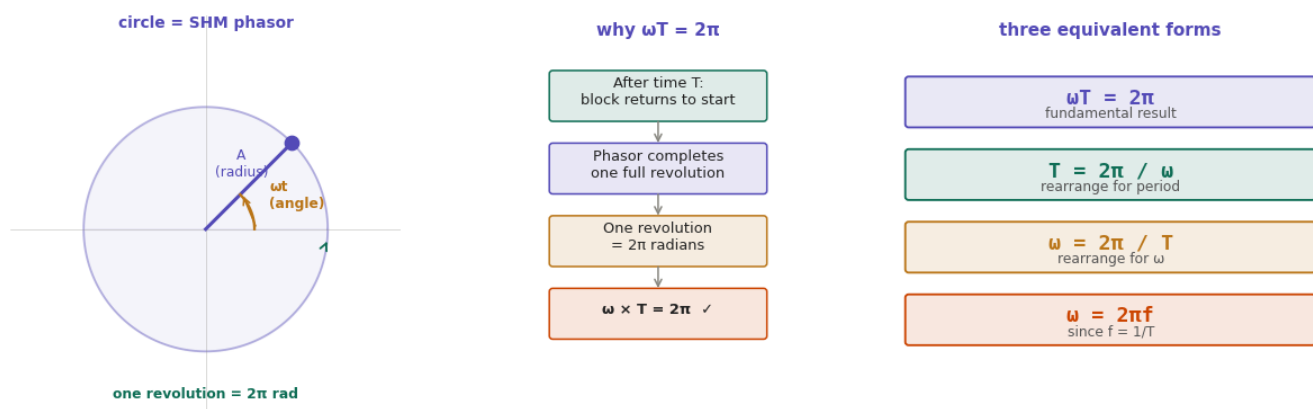
$$\omega \times T = 2\pi$$

Step 5 — Rearrange to get all three forms:

$$T = 2\pi / \omega \quad (\text{solve for period } T)$$

$$\omega = 2\pi / T \quad (\text{solve for } \omega)$$

$$\omega = 2\pi f \quad (\text{since } f = 1/T)$$



HOW IT IS READ (left panel)

The circle (phasor) has radius A (the amplitude). Point P spins at angular speed ω (rad/s). After time t the swept angle = ωt radians. The x-projection of P gives $x(t) = A \cos(\omega t)$ — exactly SHM. One full spin covers 2π radians.

THE LOGIC (middle panel)

(1) SHM repeats every $T \rightarrow$ block is back at the same position and moving the same direction. (2) On the circle, P has completed exactly one revolution. (3) One revolution = 2π radians, and angle = $\omega \times$ time, so $\omega \times T = 2\pi$. That is the derivation.

THREE EQUIVALENT FORMS (right panel)

$\omega T = 2\pi$ (fundamental) $\rightarrow T = 2\pi/\omega$ (solve for period) $\rightarrow \omega = 2\pi/T$ (solve for ω) $\rightarrow \omega = 2\pi f$ (since $f = 1/T$). All four say the same thing. The one you use depends on what you are given in a problem.

15-2 E | Velocity of SHM — Derivation

DERIVATION — Velocity $v(t)$ (Eq. 15-6)

Start: $x(t) = x_m \cos(\omega t + \phi)$

Differentiate: $v(t) = dx/dt = d/dt[x_m \cos(\omega t + \phi)]$

$= x_m \cdot (-\sin(\omega t + \phi)) \cdot \omega$ $\leftarrow$ chain rule: $d/du[\cos u] = -\sin u$, $u = \omega t + \phi$

$v(t) = -\omega x_m \sin(\omega t + \phi)$ $\leftarrow$ Eq. 15-6

Velocity amplitude: $v_m = \omega x_m$

v is maximum ($= \pm\omega x_m$) when $|\sin| = 1$, i.e. at $x = 0$ (equilibrium)
 v is zero when $\sin = 0$, i.e. at $x = \pm x_m$ (turning points)

Velocity (Eq. 15-6)

$$v(t) = -\omega x_m \sin(\omega t + \phi)$$

Velocity amplitude $v_m = \omega x_m$. Maximum at $x = 0$; zero at $x = \pm x_m$.

15-2 E(i) | Derivation 2 — Velocity Step-by-Step

Velocity is the first time derivative of displacement. Here is the step-by-step derivation:

Step	Equation / Result	Explanation
Step 1 Start with displacement	$x(t) = A \cos(\omega t + \phi)$	Our established result from Derivation 1 / Eq. 15-3.
Step 2 Apply definition $v = dx/dt$	$v(t) = dx/dt = d/dt[A \cos(\omega t + \phi)]$	Velocity is the first time derivative of displacement.
Step 3 Apply the chain rule	$v(t) = A \cdot (-\sin(\omega t + \phi)) \cdot \omega$	$d/du[\cos u] = -\sin u$; $u = \omega t + \phi$; $du/dt = \omega$. Bring ω to front.
Step 4 Final result	$v(t) = -A\omega \sin(\omega t + \phi)$	Velocity is a negative sine wave. Maximum speed $A\omega$ when $ \sin = 1$, i.e. at $x = 0$.

- **Maximum speed:** $v_m = A\omega$ — occurs at $x = 0$ (equilibrium), where all energy is kinetic.
- **At turning points $x = \pm A$:** $v = 0$ — the object momentarily stops before reversing direction.

15-2 F | Acceleration of SHM — Derivation**DERIVATION — Acceleration $a(t)$ (Eq. 15-7 & 15-8)**

Start: $v(t) = -\omega x_m \sin(\omega t + \phi)$

Differentiate: $a(t) = dv/dt = d/dt[-\omega x_m \sin(\omega t + \phi)]$

$= -\omega x_m \cdot \cos(\omega t + \phi) \cdot \omega$ ← chain rule: $d/du[\sin u] = \cos u$

$a(t) = -\omega^2 x_m \cos(\omega t + \phi)$ ← Eq. 15-7

Since $x(t) = x_m \cos(\omega t + \phi)$:

$a(t) = -\omega^2 \cdot x(t)$ ← Eq. 15-8 THE HALLMARK OF SHM

Acceleration amplitude: $a_m = \omega^2 x_m$

a is maximum at $x = \pm x_m$; a is zero at $x = 0$

Acceleration (Eq. 15-7 / 15-8)

$$a(t) = -\omega^2 x_m \cos(\omega t + \phi) = -\omega^2 x(t)$$

Acceleration amplitude $a_m = \omega^2 x_m$. Hallmark: $a \propto -x$. Always restoring. Max at $\pm x_m$; zero at 0.

HALLMARK OF SHM

$a = -\omega^2 x$. Acceleration is proportional to displacement and always opposite in sign. This single equation defines SHM. If you see this relationship in any system, that system performs SHM.

15-2 F(i) | Derivation 3 — Acceleration Step-by-Step

Acceleration is the first time derivative of velocity. Here is the full step-by-step derivation:

Step	Equation / Result	Explanation
Step 1 Start with velocity	$v(t) = -A\omega \sin(\omega t + \phi)$	Our result from Derivation 2 / Eq. 15-6.
Step 2 Apply definition $a = dv/dt$	$a(t) = dv/dt = d/dt[-A\omega \sin(\omega t + \phi)]$	Acceleration is the first derivative of velocity (second derivative of displacement).
Step 3 Apply chain rule	$a(t) = -A\omega \cdot \cos(\omega t + \phi) \cdot \omega$	$d/du[\sin u] = \cos u$; $du/dt = \omega$. Factor brings another ω to the front.
Step 4 Collect terms	$a(t) = -A\omega^2 \cos(\omega t + \phi)$	Group the two ω factors into ω^2 .
Step 5 Recognise $x(t)$	$a(t) = -\omega^2 \cdot [A \cos(\omega t + \phi)] = -\omega^2 x$	The bracket is exactly $x(t)$. So $a = -\omega^2 x$ — this is the defining SHM condition, confirmed. ✓

RECALL — Why $a = d^2x/dt^2$

$$x(t) \rightarrow v = dx/dt \rightarrow a = dv/dt = d^2x/dt^2$$

Velocity is the 1st derivative of position; acceleration is the 1st derivative of velocity, so it is the 2nd derivative of position. Differentiating $x(t) = A \cos(\omega t)$ twice gives $a = -A\omega^2 \cos(\omega t) = -\omega^2 x$, confirming Step 5 above. See the full explanation in Section 15-2 C(i).

- **Maximum acceleration:** $a_m = A\omega^2$ — at turning points $x = \pm A$, where restoring force is maximum.
- **At equilibrium ($x = 0$):** $a = 0$ — the net force is zero there ($F = -kx = 0$, spring is relaxed).

15-2 G | Phase Relationships — x , v , and a

x , v , and a all oscillate at the same frequency ω but with different phase shifts. Understanding these shifts is crucial:

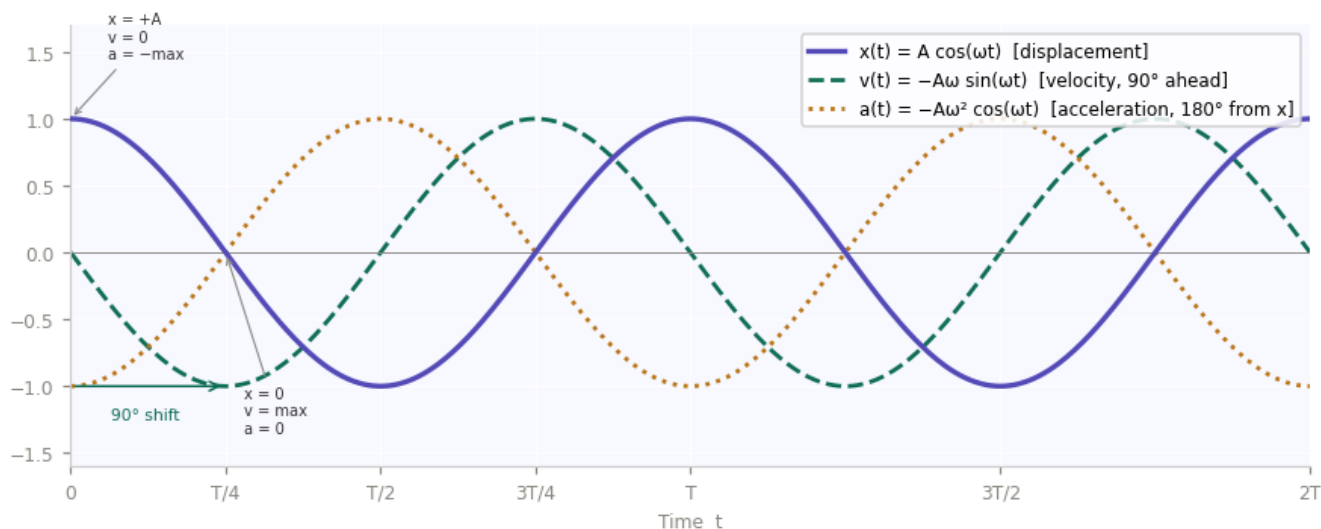
PHASE RELATIONSHIPS EXPLAINED INTUITIVELY

x and v are 90° out of phase: When displacement is maximum (turning point), velocity is zero — mass has stopped to reverse direction. When displacement is zero (equilibrium), velocity is maximum — mass flies through at full speed.

x and a are 180° out of phase: Acceleration always opposes displacement. Displaced right ($+x$) $\rightarrow$ pulled left ($-a$). Displaced left ($-x$) $\rightarrow$ pushed right ($+a$). This is exactly what $F = -kx$ means.

v and a are 90° out of phase: Acceleration peaks when velocity is zero (at turning points — maximum restoring force). Acceleration is zero when velocity is maximum (at equilibrium — spring is relaxed, no force).

Quantity	Max value	Occurs at	Formula
Displacement x	x_m	$x = \pm x_m$ (turning points)	$x = x_m \cos(\omega t + \phi)$
Velocity v	$v_m = \omega x_m$	$x = 0$ (equilibrium)	$v = -\omega x_m \sin(\omega t + \phi)$
Acceleration a	$a_m = \omega^2 x_m$	$x = \pm x_m$ (turning points)	$a = -\omega^2 x_m \cos(\omega t + \phi)$
Force F	$F_m = k x_m$	$x = \pm x_m$ (turning points)	$F = -kx$



Graph 2 — Phase relationships: $x(t)$, $v(t)$, and $a(t)$ on the same axis

Purple solid: $x(t) = A \cos(\omega t)$. Green dashed: $v(t) = -A\omega \sin(\omega t)$. Amber dotted: $a(t) = -A\omega^2 \cos(\omega t)$.

v is 90° ahead of x — when x peaks, v is already at zero; when $x = 0$, v has just peaked.

a is 180° opposite to x — when x is most positive, a is most negative (spring pulling hardest back).

Key check at $T/4$: $x = 0$, $v = \text{maximum}$, $a = 0$. At $T/2$: $x = -A$, $v = 0$, $a = +\text{maximum}$.

This 180° relationship ($a = -\omega^2 x$) is THE defining equation of SHM — it appears in every problem.

15-2 H | Quick Reference Table

Position	x	v	a	F
Equilibrium $x=0$	0	Max = $\pm \omega x_m$	0	0
Amplitude $x=+x_m$	$+x_m$	0	Max = $-\omega^2 x_m$	$-k x_m$
Amplitude $x=-x_m$	$-x_m$	0	Max = $+\omega^2 x_m$	$+k x_m$

15-2 I | Phase Constant ϕ — Initial Conditions

The phase constant ϕ is fixed by the state of the system at $t = 0$:

At $t = 0$: $x(0) = x_m \cos(\phi)$ → gives $\cos(\phi)$

At $t = 0$: $v(0) = -\omega x_m \sin(\phi)$ → gives $\sin(\phi)$

Dividing: $\tan(\phi) = -v(0) / [\omega \cdot x(0)]$

Amplitude from initial conditions: $x_m = \sqrt{x_0^2 + (v_0/\omega)^2}$

Special cases:

$\phi = 0 \rightarrow x(0) = x_m, v(0) = 0$ (released from rest at maximum)

$\phi = \pi \rightarrow x(0) = -x_m, v(0) = 0$ (released from rest at minimum)

$\phi = \pi/2 \rightarrow x(0) = 0, v(0) = -\omega x_m$ (at equilibrium, moving in -x direction)

The Force Law for Simple Harmonic Motion

Hooke's Law, ODE derivation, spring constant, angular frequency, period, and isochronism

Knowing $a = -\omega^2 x$, we apply Newton's Second Law $F = ma$ to find the force producing SHM. This links the kinematics of SHM to its physical cause.

15-3 A | Hooke's Law — The Foundation

Hooke's Law $F = -kx$ (Eq. 15-10)

$$\mathbf{F} = -k\mathbf{x}$$

F = restoring force (N); k = spring constant (N/m, always positive); x = displacement from equilibrium (m). The minus sign makes F always oppose displacement — it is a restoring force.

Symbol	Physical meaning	Sign convention	Effect
F	Restoring force (N)	$+$ = rightward/upward	Determines acceleration direction
k	Spring constant (N/m)	Always positive	Larger $k \rightarrow$ stronger force $\rightarrow$ faster oscillation
x	Displacement from equilib. (m)	$+$ = right/away from equilibrium	Determines force magnitude
$-$ sign	Makes F oppose x	F and x always opposite	THE mechanism of self-correction

Physical meaning of k — examples:

Spring	k value	What it means
Gentle/soft spring	~ 10 N/m	10 N to stretch/compress 1 metre. Slow oscillation.
Car suspension spring	15 000–25 000 N/m	Very stiff. Handles large forces at highway speeds.
Atomic bond (approx.)	~ 0.01 N/m	The 'spring' between two bonded atoms. Microscopic scale.

SELF-CORRECTING MECHANISM

No matter which direction the mass is displaced, $F = -kx$ always pushes it back toward equilibrium:

Displaced RIGHT ($x > 0$): $F = -k(+x) = \text{negative} \rightarrow$ force points LEFT (restoring $\checkmark$)

At equilibrium ($x = 0$): $F = 0 \rightarrow$ no net force (resting $\checkmark$)

Displaced LEFT ($x < 0$): $F = -k(-x) = \text{positive} \rightarrow \text{force points RIGHT (restoring)}$ ✓

Common Mistake: Writing $F = kx$ (forgetting the minus sign). Without the minus sign the equation describes exponential growth, not oscillation. The minus sign is not optional — it is the mathematical statement that force opposes displacement.

15-3 B | Full ODE Derivation — From Hooke's Law to $x(t)$

DERIVATION — SHM Differential Equation & Solution (Eq. 15-9 through 15-13)

Step 1 — Apply Newton's Second Law:

$$F_{\text{net}} = m \cdot a$$

Step 2 — The only force is the spring (Hooke's Law):

$$-kx = m \cdot a$$

Step 3 — Write acceleration as second derivative of position:

$$a = d^2x/dt^2$$

Step 4 — Substitute into the equation:

$$m \cdot (d^2x/dt^2) = -kx$$

Step 5 — Divide both sides by m :

$$d^2x/dt^2 = -(k/m) \cdot x$$

Step 6 — Define $\omega^2 = k/m$ (angular frequency squared):

$$d^2x/dt^2 + \omega^2 x = 0 \quad \leftarrow \text{THE SHM DIFFERENTIAL EQUATION}$$

Step 7 — Propose trial solution $x(t) = A \cos(\omega t + \phi)$:

$$dx/dt = -A\omega \sin(\omega t + \phi)$$

$$d^2x/dt^2 = -A\omega^2 \cos(\omega t + \phi) = -\omega^2 \cdot [A \cos(\omega t + \phi)] = -\omega^2 x \quad \checkmark$$

Step 8 — Substitute into ODE to verify:

$$-\omega^2 x + \omega^2 x = 0 \quad \checkmark \quad \text{The solution is confirmed.}$$

Result: $x(t) = A \cos(\omega t + \phi)$ with $\omega = \sqrt{k/m}$

A and ϕ are determined by initial conditions x_0 and v_0 .

Force Law for SHM (Eq. 15-9)

$$\mathbf{F} = -m\omega^2 \mathbf{x} = -k\mathbf{x}$$

Restoring force $\propto -x$. This is Hooke's Law. The spring constant $k = m\omega^2$.

Spring Constant & Angular Frequency (Eq. 15-11 / 15-12)

$$k = m\omega^2 \quad \rightarrow \quad \omega = \sqrt{k/m}$$

ω depends only on k (springiness) and m (inertia). Stiffer spring $\rightarrow$ higher ω . Heavier mass $\rightarrow$ lower ω .

Period — Spring-Block System (Eq. 15-13)

$$T = 2\pi \sqrt{m/k}$$

Period depends only on m and k — NOT on amplitude x_m . This is isochronism.

15-3 C | Isochronism — Why Period is Independent of Amplitude

ISOCHRONISM — The Most Remarkable Property of SHM

The period $T = 2\pi\sqrt{m/k}$ contains NO amplitude A . A mass pulled 1 cm oscillates with exactly the same period as one pulled 10 cm. This is because larger amplitude means more distance to travel, but also proportionally stronger restoring force — the two effects cancel perfectly. It is the reason mechanical clocks keep accurate time regardless of initial displacement.

Change made	Effect on ω	Effect on T	Physical reason
Double mass $m \rightarrow 2m$	$\omega \rightarrow \omega/\sqrt{2}$ (decreases)	$T \rightarrow \sqrt{2} \cdot T$ (increases)	More inertia resists acceleration
Double spring $k \rightarrow 2k$	$\omega \rightarrow \sqrt{2} \cdot \omega$ (increases)	$T \rightarrow T/\sqrt{2}$ (decreases)	Stronger restoring force
Double amplitude $A \rightarrow 2A$	No change	No change	Isochronism: T independent of A
Move to Moon ($g/6$)	No change ($\omega = \sqrt{k/m}$)	No change	Spring period does not involve g

15-3 D | Physical Interpretation

SPRINGINESS vs INERTIA

Every SHM system needs two ingredients: (1) **SPRINGINESS** — a restoring force (spring constant k) that pulls the system back to equilibrium and stores potential energy. (2) **INERTIA** — mass m that resists changes in velocity and stores kinetic energy. They compete: more springiness speeds up the oscillation; more inertia slows it down. $T = 2\pi\sqrt{m/k}$ captures this competition exactly.

15-3 E | Checkpoint 2

Which of the following force–position relationships implies SHM?

(a) $F = -5x$ (b) $F = -400x^2$ (c) $F = 10x$ (d) $F = 3x^2$

Answer: (a) only. $F = -5x$ has $F \propto -x$ (linear, restoring) $\rightarrow$ SHM with $k = 5$ N/m.

Why the others fail: (b) $F = -400x^2$ is nonlinear (x squared, not x); (c) $F = +10x$ has the wrong sign — force is in the same direction as displacement, so it drives the mass away from equilibrium rather than restoring it; (d) $F = 3x^2$ is both nonlinear and the wrong sign.

Energy in Simple Harmonic Motion

Potential energy, kinetic energy, total mechanical energy, v-x relationship, and energy graphs

In a frictionless SHM system, total mechanical energy is perfectly conserved. Energy continuously converts between kinetic energy K and elastic potential energy U , but their sum E never changes. We now derive each quantity and explore all implications.

15-4 A | Potential Energy — Derivation

The potential energy is stored entirely in the spring. From mechanics, the work done stretching a spring by x from rest is the integral of F over displacement:

DERIVATION — Potential Energy $U(t)$ (Eq. 15-18)

Work done compressing/stretching spring by x :

$$W = \int_0^x k \cdot x \, dx = \frac{1}{2}kx^2 \quad \rightarrow \text{this work is stored as PE}$$

$$U = \frac{1}{2}kx^2$$

Substitute $x(t) = x_m \cos(\omega t + \phi)$:

$$U(t) = \frac{1}{2}k [x_m \cos(\omega t + \phi)]^2$$

$$U(t) = \frac{1}{2}k x_m^2 \cos^2(\omega t + \phi) \quad \leftarrow \text{Eq. 15-18}$$

$$\text{Also: } U = \frac{1}{2}m\omega^2 x_m^2 \cos^2(\omega t + \phi) \quad (\text{using } k = m\omega^2)$$

U is maximum ($= \frac{1}{2}kx_m^2$) when $\cos^2 = 1$, i.e. at $x = \pm x_m$

U is zero when $\cos^2 = 0$, i.e. at $x = 0$ (equilibrium)

U is always ≥ 0 (parabola — never negative)

Potential Energy $U(t)$ (Eq. 15-18)

$$U(t) = \frac{1}{2}kx^2 = \frac{1}{2}kx_m^2 \cos^2(\omega t + \phi)$$

Stored in the spring. Maximum $\frac{1}{2}kx_m^2$ at amplitude; zero at equilibrium. Oscillates at angular frequency 2ω .

CAUTION — Notation

$\cos^2 A$ means $(\cos A)^2$, NOT $\cos(A^2)$. For example: $\cos^2(90^\circ) = (\cos 90^\circ)^2 = 0^2 = 0$. But $\cos(90^{\circ 2}) = \cos(8100^\circ)$ which is not zero. These are completely different functions!

15-4 B | Kinetic Energy — Derivation

DERIVATION — Kinetic Energy $K(t)$ (Eq. 15-19 / 15-20)

Kinetic energy: $K = \frac{1}{2}mv^2$

Substitute $v(t) = -\omega x_m \sin(\omega t + \phi)$:

$$K(t) = \frac{1}{2}m [-\omega x_m \sin(\omega t + \phi)]^2$$

$$K(t) = \frac{1}{2}m \omega^2 x_m^2 \sin^2(\omega t + \phi) \quad \leftarrow \text{Eq. 15-19}$$

Use $k = m\omega^2$ (from Eq. 15-11):

$$K(t) = \frac{1}{2}kxm^2 \sin^2(\omega t + \phi) \quad \leftarrow \text{Eq. 15-20}$$

K is maximum ($= \frac{1}{2}kxm^2$) when $\sin^2 = 1$, i.e. at $x = 0$

K is zero when $\sin^2 = 0$, i.e. at $x = \pm x_m$ (turning points)

K is always ≥ 0 (inverted parabola — never negative)

Kinetic Energy $K(t)$ (Eq. 15-19 / 15-20)

$$K(t) = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 x_m^2 \sin^2(\omega t + \phi) = \frac{1}{2}kxm^2 \sin^2(\omega t + \phi)$$

Carried by the moving block. Maximum $\frac{1}{2}kxm^2$ at equilibrium; zero at amplitude. Also oscillates at 2ω .

15-4 C | Total Mechanical Energy — Derivation

DERIVATION — Total Mechanical Energy E (Eq. 15-21)

$$\begin{aligned} E &= U + K \\ &= \frac{1}{2}kxm^2 \cos^2(\omega t + \phi) + \frac{1}{2}kxm^2 \sin^2(\omega t + \phi) \\ &= \frac{1}{2}kxm^2 [\cos^2(\omega t + \phi) + \sin^2(\omega t + \phi)] \end{aligned}$$

Apply the Pythagorean identity: $\cos^2\alpha + \sin^2\alpha = 1$ (for any angle α)

$$E = \frac{1}{2}kxm^2 \cdot [1]$$

$$E = \frac{1}{2}kxm^2 \quad \leftarrow \text{Eq. 15-21} \quad (\text{CONSTANT!})$$

Proof that KE and PE oscillate at 2ω :

$$KE(t) = \frac{1}{2}kxm^2 \cos^2(\omega t) = \frac{1}{2}kxm^2 \cdot [1 + \cos(2\omega t)]/2$$

$$PE(t) = \frac{1}{2}kxm^2 \sin^2(\omega t) = \frac{1}{2}kxm^2 \cdot [1 - \cos(2\omega t)]/2$$

The $\cos(2\omega t)$ term oscillates at twice the SHM frequency 2ω .

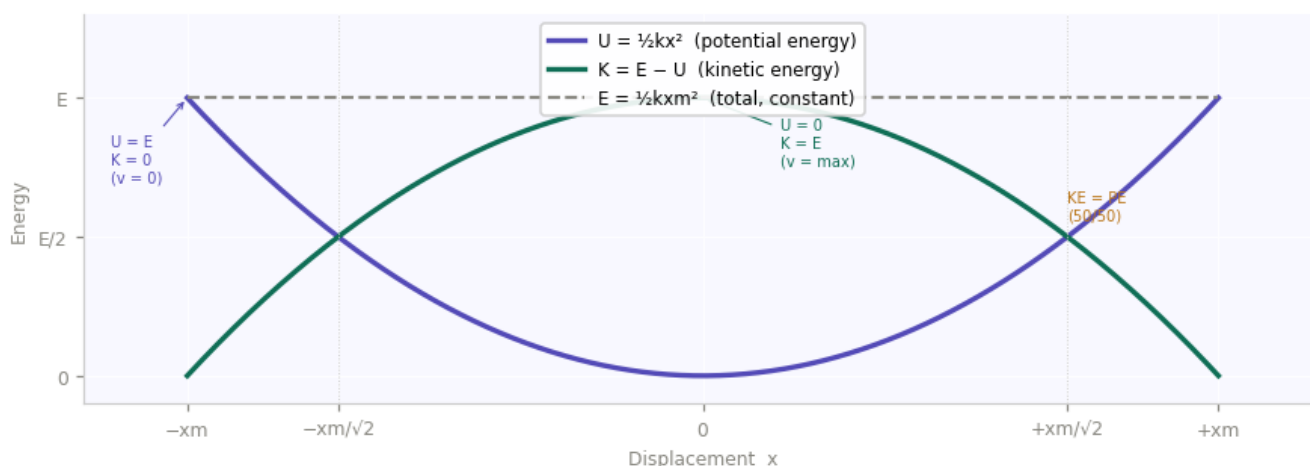
Total Mechanical Energy (CONSTANT) (Eq. 15-21)

$$E = U + K = \frac{1}{2}kxm^2$$

E is constant — independent of time, position, velocity, or phase constant. Depends only on spring constant k and amplitude x_m .

WHY E IS CONSERVED — AND WHY KE AND PE OSCILLATE AT 2ω

As block moves toward equilibrium: spring force does positive work, converting $U \rightarrow K$. Moving away: block does work against spring, converting $K \rightarrow U$. Sum always $\frac{1}{2}kxm^2$. The $\cos^2/\sin^2$ functions both complete TWO full oscillation cycles per SHM cycle — this is why energy oscillates at 2ω . Average values: $\langle KE \rangle = \langle PE \rangle = \frac{1}{4}kxm^2$ over one full period.



Graph 3 — Energy vs position: PE and KE parabolas

Purple upward parabola: $U = \frac{1}{2}kx^2$. Grows from zero at $x = 0$ to maximum $\frac{1}{2}kxm^2$ at $x = \pm xm$.

Teal inverted parabola: $K = E - U = \frac{1}{2}k(xm^2 - x^2)$. Maximum at $x = 0$, falls to zero at $\pm xm$.

Gray dashed line: total energy $E = \frac{1}{2}kxm^2$ — perfectly constant at every position.

The two parabolas are mirror images — their sum is always the same constant E .

Dotted vertical lines at $x = \pm xm/\sqrt{2}$: here $KE = PE$ exactly (50-50 energy split point).

Formula from graph: $v = \omega\sqrt{(xm^2 - x^2)}$ — speed at any position without needing time.

15-4 D | Energy at Key Positions

Position	Potential U	Kinetic K	Speed v	Total E
$x = +xm$ (amplitude)	$\frac{1}{2}kxm^2$ (max)	0	0 (stopped)	$\frac{1}{2}kxm^2$
$x = +xm/2$	$\frac{1}{4} \cdot (\frac{1}{2}kxm^2)$	$\frac{3}{4} \cdot (\frac{1}{2}kxm^2)$	$vm \cdot (\sqrt{3}/2)$	$\frac{1}{2}kxm^2$
$x = 0$ (equilibrium)	0	$\frac{1}{2}kxm^2$ (max)	$vm = \omega xm$	$\frac{1}{2}kxm^2$
$x = -xm/2$	$\frac{1}{4} \cdot (\frac{1}{2}kxm^2)$	$\frac{3}{4} \cdot (\frac{1}{2}kxm^2)$	$vm \cdot (\sqrt{3}/2)$	$\frac{1}{2}kxm^2$
$x = -xm$ (amplitude)	$\frac{1}{2}kxm^2$ (max)	0	0 (stopped)	$\frac{1}{2}kxm^2$

EQUAL ENERGY SPLIT AT $x = \pm xm/\sqrt{2}$

$KE = PE$ when $\frac{1}{2}kxm^2 \sin^2\theta = \frac{1}{2}kxm^2 \cos^2\theta \rightarrow \tan^2\theta = 1 \rightarrow x = \pm xm/\sqrt{2} \approx \pm 0.707 \cdot xm$. At this position each form holds exactly half the total energy ($\frac{1}{4}kxm^2$ each). This is the 50-50 energy split point, located at about 70.7% of the amplitude.

15-4 E | Velocity from Energy — the v-x Relationship

We can find speed at any displacement x directly from energy conservation, without needing the time equations:

DERIVATION — $v(x)$: Velocity from Energy Conservation

Total energy: $E = U + K = \frac{1}{2}kxm^2$

Expand: $\frac{1}{2}kxm^2 = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$

$$\text{Rearrange: } \frac{1}{2}mv^2 = \frac{1}{2}kxm^2 - \frac{1}{2}kx^2 = \frac{1}{2}k(xm^2 - x^2)$$

$$v^2 = (k/m)(xm^2 - x^2) = \omega^2(xm^2 - x^2)$$

$$v = \pm\omega\sqrt{(xm^2 - x^2)} \quad \leftarrow \text{speed at any position } x$$

Check at $x = 0$: $v = \pm\omega \cdot xm = \pm v_m$ ✓ (maximum speed)

Check at $x = \pm xm$: $v = 0$ ✓ (turning points, mass stopped)

The v - x curve is a SEMI-ELLIPSE:

$$\text{Rearrange: } v^2/(\omega xm)^2 + x^2/xm^2 = 1$$

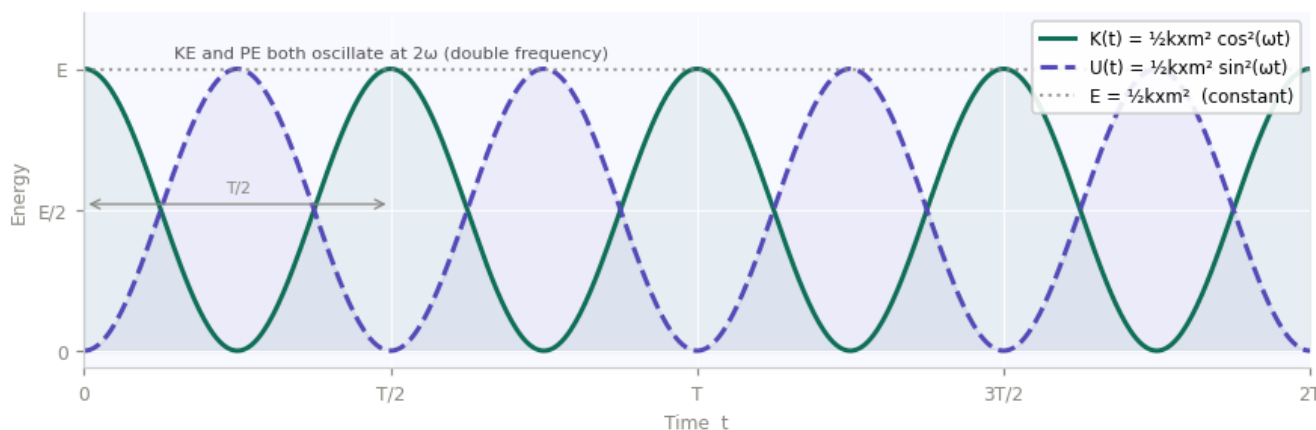
This is the equation of an ellipse in x - v phase space!

Semi-axes: xm (horizontal), $\omega xm = v_m$ (vertical)

Speed at Any Position x

$$v = \pm\omega \sqrt{(xm^2 - x^2)}$$

No trigonometry needed — velocity at any displacement from energy alone. The v - x curve is a semi-ellipse with semi-axes xm and $v_m = \omega xm$.



Graph 4 — Energy vs time: KE and PE oscillating at double frequency 2ω

Teal: $K(t) = \frac{1}{2}kxm^2 \cos^2(\omega t)$. Starts maximum (block at $x = 0$ moving fast) when $\phi = \pi/2$.

Purple dashed: $U(t) = \frac{1}{2}kxm^2 \sin^2(\omega t)$. Starts at zero when $\phi = 0$, rises as block moves away.

Both curves oscillate at frequency 2ω — twice the SHM frequency. This is because $\cos^2/\sin^2$ complete two cycles per SHM cycle.

They are always exactly out of phase: when K is maximum, U is zero, and vice versa.

Their sum equals the constant gray line $E = \frac{1}{2}kxm^2$ at every instant (Pythagorean identity: $\cos^2 + \sin^2 = 1$).

15-4 F | Checkpoint 3

$K = 3 \text{ J}$ and $U = 2 \text{ J}$ when the block is at $x = +2.0 \text{ cm}$.

(a) What is K at $x = 0$?

Answer: $E = K + U = 3 + 2 = 5 \text{ J}$. At $x = 0$, $U = \frac{1}{2}k(0)^2 = 0$. So $K = E = 5 \text{ J}$. All energy is kinetic; block moves at maximum speed.

(b) What is U at $x = -2.0$ cm?

Answer: $U = \frac{1}{2}kx^2$ depends on $|x|^2$ only, not the sign of x . At $x = -2.0$ cm, spring stretch magnitude is identical to at $+2.0$ cm. Therefore $U = 2$ J (unchanged).

(c) What is U at $x = -x_m$?

Answer: At $x = \pm x_m$ the block is momentarily at rest, so $K = 0$. Therefore $U = E - 0 = 5$ J. All mechanical energy is stored in the spring.

Sample Problem**SHM — Period, Frequency, Angular Frequency, Maximum Speed**

Given: $m = 0.5$ kg, $k = 200$ N/m, $A = 0.10$ m. Find: (a) ω (b) T (c) f (d) v_m

KEY IDEA

The block–spring system is a linear SHM oscillator. Use $\omega = \sqrt{k/m}$, then $T = 2\pi/\omega$, $f = 1/T$, $v_m = A\omega$.

(a) Angular frequency ω

$$\omega = \sqrt{k/m} = \sqrt{(200/0.5)} = \sqrt{400} = 20 \text{ rad/s}$$

(b) Period T

$$T = 2\pi/\omega = 2\pi/20 = 0.314 \text{ s} \approx 314 \text{ ms}$$

(c) Frequency f

$$f = 1/T = 1/0.314 = 3.18 \text{ Hz}$$

(d) Maximum speed v_m

$$v_m = A\omega = (0.10 \text{ m})(20 \text{ rad/s}) = 2.0 \text{ m/s} \quad (\text{occurs at } x = 0)$$

Sample Problem**SHM — Energy at a Given Position**

Given: $m = 0.4$ kg, $k = 100$ N/m, $A = 0.15$ m, $x = 0.09$ m. Find: E_{total} , PE, KE, and speed v at $x = 0.09$ m.

KEY IDEA

Use $E = \frac{1}{2}kA^2$, $PE = \frac{1}{2}kx^2$, $KE = E - PE$, then $v = \sqrt{(2KE/m)}$.

$$E_{\text{total}} = \frac{1}{2}kA^2 = \frac{1}{2}(100)(0.15)^2 = \frac{1}{2}(100)(0.0225) = 1.125 \text{ J}$$

$$PE = \frac{1}{2}kx^2 = \frac{1}{2}(100)(0.09)^2 = \frac{1}{2}(100)(0.0081) = 0.405 \text{ J}$$

$$KE = E - PE = 1.125 - 0.405 = 0.720 \text{ J}$$

$$v = \sqrt{(2 \cdot KE/m)} = \sqrt{(2 \times 0.720/0.4)} = \sqrt{3.6} \approx 1.90 \text{ m/s}$$

Check using $v = \omega \sqrt{(A^2 - x^2)}$:

$$\omega = \sqrt{(k/m)} = \sqrt{(100/0.4)} = \sqrt{250} \approx 15.81 \text{ rad/s}$$

$$v = 15.81 \cdot \sqrt{(0.0225 - 0.0081)} = 15.81 \cdot \sqrt{0.0144} = 15.81 \times 0.12 \approx 1.90 \text{ m/s} \quad \checkmark$$

Sample Problem

SHM Potential Energy, Kinetic Energy — Mass Dampers

A tall-building mass damper: $m = 2.72 \times 10^5 \text{ kg}$, $f = 10.0 \text{ Hz}$, $x_m = 20.0 \text{ cm}$. Find (a) E and (b) speed at $x = 0$.

KEY IDEA

E is constant throughout motion. Evaluate at $x = x_m$ where $v = 0$, so $E = U$ only. At $x = 0$, $E = K$ only.

(a) Total mechanical energy E

$$\omega = 2\pi f = 2\pi(10.0) = 62.83 \text{ rad/s}$$

$$k = m\omega^2 = (2.72 \times 10^5)(62.83)^2 = (2.72 \times 10^5)(3947.6) = 1.073 \times 10^9 \text{ N/m}$$

$$E = \frac{1}{2}kx_m^2 = \frac{1}{2}(1.073 \times 10^9)(0.20)^2 = \frac{1}{2}(1.073 \times 10^9)(0.04) = 2.15 \times 10^7 \text{ J}$$

$$\approx 2.1 \times 10^7 \text{ J}$$

(b) Speed at $x = 0$ (equilibrium)

$$\text{At } x = 0: U = 0, \text{ so } E = K = \frac{1}{2}mv^2$$

$$2.15 \times 10^7 = \frac{1}{2}(2.72 \times 10^5)v^2$$

$$v^2 = (2 \times 2.15 \times 10^7) / (2.72 \times 10^5) = 158.1 \text{ m}^2/\text{s}^2$$

$$v = 12.6 \text{ m/s} \quad (= v_m, \text{ the maximum speed})$$

Sample Problem

Block-Spring SHM: Full Analysis — ω , T , x_m , v_m , a_m , ϕ , $x(t)$

$m = 680 \text{ g}$, $k = 65 \text{ N/m}$, block pulled $x = 11 \text{ cm}$ and released from rest at $t = 0$.

KEY IDEA

Released from rest at maximum displacement $\rightarrow$ amplitude = 11 cm. Phase constant from initial conditions.

$$(a) \quad \omega = \sqrt{(k/m)} = \sqrt{(65/0.68)} = \sqrt{95.59} = 9.78 \text{ rad/s}$$

- (b) $f = \omega/2\pi = 9.78/6.283 = 1.56 \text{ Hz}$
- (c) $T = 1/f = 0.64 \text{ s} = 640 \text{ ms}$
- (d) $x_m = 11 \text{ cm} = 0.11 \text{ m}$ (released from rest at amplitude)
- (e) $v_m = \omega x_m = (9.78)(0.11) = 1.08 \text{ m/s}$ at $x = 0$
- (f) $a_m = \omega^2 x_m = (9.78)^2 (0.11) = 10.5 \text{ m/s}^2$ at $x = \pm x_m$
- (g) ϕ : $x(0) = x_m \cos(\phi) = x_m \rightarrow \cos(\phi) = 1 \rightarrow \phi = 0 \text{ rad}$
- (h) $x(t) = (0.11 \text{ m}) \cos(9.78t)$ [x in metres, t in seconds]

PRACTICE PROBLEMS — with Complete Solutions

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Practice Problem 1 | Restoring Force and Acceleration at Multiple Positions

Given: $k = 80 \text{ N/m}$, $m = 0.5 \text{ kg}$, $A = 0.25 \text{ m}$. Find F and a at: (a) $x = 0$ (b) $x = 0.10 \text{ m}$ (c) $x = A$

Preliminary:

$$\omega^2 = k/m = 80/0.5 = 160 \text{ rad}^2/\text{s}^2$$

(a) $x = 0$: $F = -kx = -80(0) = 0 \text{ N}$. $a = F/m = 0 \text{ m/s}^2$. Equilibrium — spring relaxed, no force.

(b) $x = 0.10 \text{ m}$: $F = -kx = -80(0.10) = -8.0 \text{ N}$ (toward equilibrium). $a = F/m = -8.0/0.5 = -16 \text{ m/s}^2$.

(c) $x = A = 0.25 \text{ m}$: $F = -kx = -80(0.25) = -20 \text{ N}$. $a = -20/0.5 = -40 \text{ m/s}^2$. Maximum force and acceleration.

Practice Problem 2 | Complete Energy Analysis

Given: $m = 0.20 \text{ kg}$, $k = 80 \text{ N/m}$, $A = 0.10 \text{ m}$. Find: ω , T , f , E , v_m , KE and PE at $x = 0.06 \text{ m}$, and x where $KE = PE$.

ω : $\sqrt{k/m} = \sqrt{80/0.20} = \sqrt{400} = 20 \text{ rad/s}$

T : $2\pi/\omega = 2\pi/20 = 0.314 \text{ s}$ $f = 1/T = 3.18 \text{ Hz}$

E : $\frac{1}{2}kA^2 = \frac{1}{2}(80)(0.10)^2 = \frac{1}{2}(80)(0.01) = 0.40 \text{ J}$

v_m : $A\omega = (0.10)(20) = 2.0 \text{ m/s}$ (at $x = 0$)

At $x = 0.06 \text{ m}$:

$$PE = \frac{1}{2}kx^2 = \frac{1}{2}(80)(0.06)^2 = \frac{1}{2}(80)(0.0036) = 0.144 \text{ J}$$

$$KE = E - PE = 0.40 - 0.144 = 0.256 \text{ J}$$

$KE = PE$ at: $\frac{1}{2}kxm^2 \sin^2\theta = \frac{1}{2}kxm^2 \cos^2\theta \rightarrow x = \pm A/\sqrt{2} = \pm 0.10/\sqrt{2} \approx \pm 0.0707 \text{ m}$ (50-50 energy split)

Practice Problem 3 | Velocity–Displacement Relationship

Given: $A = 0.30 \text{ m}$, $\omega = 10 \text{ rad/s}$. Find: (a) v at $x = 0.20 \text{ m}$ (b) x when $v = 2.0 \text{ m/s}$ (c) v_m and a_m (d) KE/E ratio at $x = 0.20 \text{ m}$

(a) v at $x = 0.20 \text{ m}$: $v = \omega\sqrt{A^2 - x^2} = 10\sqrt{(0.09 - 0.04)} = 10\sqrt{0.05} = 10(0.2236) \approx 2.24 \text{ m/s}$

(b) x when $v = 2.0 \text{ m/s}$: $v^2 = \omega^2(A^2 - x^2) \rightarrow x^2 = A^2 - v^2/\omega^2 = 0.09 - 4/100 = 0.05 \rightarrow x = \pm 0.224 \text{ m}$

(c) v_m and a_m : $v_m = A\omega = (0.30)(10) = 3.0 \text{ m/s}$ $a_m = A\omega^2 = (0.30)(100) = 30 \text{ m/s}^2$

(d) KE/E at $x = 0.20$: $KE/E = (A^2 - x^2)/A^2 = (0.09 - 0.04)/0.09 = 0.05/0.09 = 5/9 \approx 55.6\%$

COMPLETE FORMULA REFERENCE

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Table 1 | Kinematics of SHM

Quantity	Formula	Key point
Displacement	$x(t) = x_m \cos(\omega t + \phi)$	Cosine curve; A and ϕ from initial conditions
Velocity	$v(t) = -\omega x_m \sin(\omega t + \phi)$	90° ahead of x; $v_m = \omega x_m$ at $x = 0$
Acceleration	$a(t) = -\omega^2 x_m \cos(\omega t + \phi) = -\omega^2 x$	180° from x; defines SHM; $a_m = \omega^2 x_m$
Maximum speed	$v_m = \omega x_m$	At $x = 0$ (equilibrium)
Maximum acceleration	$a_m = \omega^2 x_m$	At $x = \pm x_m$ (turning points)
Speed vs position	$v = \omega \sqrt{x_m^2 - x^2}$	Energy-based; semi-ellipse in x-v space
Amplitude from ICs	$x_m = \sqrt{x_0^2 + (v_0/\omega)^2}$	x_0, v_0 = initial position, velocity

Table 2 | Force Law and Frequency

Quantity	Formula	Key point
Restoring force	$F = -kx$	Always toward equilibrium (Hooke's Law)
Angular frequency	$\omega = \sqrt{k/m}$	Larger k or smaller m → bigger ω
Period (spring-mass)	$T = 2\pi\sqrt{m/k}$	INDEPENDENT of amplitude — isochronism!
Frequency (spring-mass)	$f = (1/2\pi)\sqrt{k/m}$	Hz = cycles per second

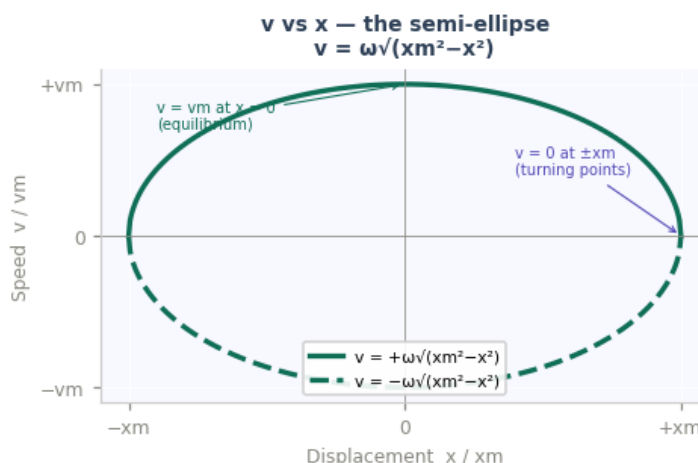
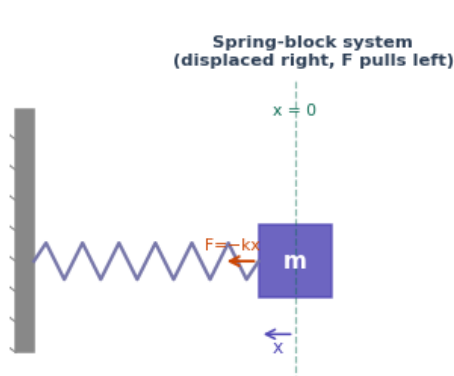
Table 3 | Energy in SHM

Quantity	Formula	Key point
Potential energy	$PE = U = \frac{1}{2}kx^2$	Upward parabola; max at $x = \pm x_m$; zero at $x = 0$
Kinetic energy	$KE = K = \frac{1}{2}k(x_m^2 - x^2)$	Inverted parabola; max at $x = 0$; zero at $x = \pm x_m$

Total energy	$E = \frac{1}{2}kx_m^2 = \text{constant}$	Conserved; proportional to x_m^2
PE as function of time	$PE(t) = \frac{1}{2}kx_m^2 \sin^2(\omega t)$	Oscillates at 2ω ; starts at 0 if $\phi = 0$
KE as function of time	$KE(t) = \frac{1}{2}kx_m^2 \cos^2(\omega t)$	Oscillates at 2ω ; starts at max if $\phi = 0$
Equal energy split	$KE = PE$ at $x = \pm x_m/\sqrt{2}$	50-50 split at 70.7% of amplitude
Average energies	$\langle KE \rangle = \langle PE \rangle = \frac{1}{4}kx_m^2$	Each is half of total energy on average

FINAL SUMMARY — THE BIG PICTURE OF SHM

SHM arises whenever $F = -kx$. The displacement is sinusoidal: $x = x_m \cos(\omega t + \phi)$. Angular frequency $\omega = \sqrt{k/m}$ depends only on stiffness and mass — not amplitude (isochronism). Velocity leads displacement by 90° ; acceleration is 180° opposite. Energy oscillates between kinetic ($\frac{1}{2}mv^2$) and potential ($\frac{1}{2}kx^2$) at double frequency, with constant total $E = \frac{1}{2}kx_m^2$.



Graph 5 — Spring-block schematic and velocity vs displacement (v-x semi-ellipse)

Left: the spring-block system. Wall anchors the spring. Spring exerts $F = -kx$ on the block (mass m). Displacement x is measured from equilibrium (dashed green line).

When block moves right ($+x$): spring is stretched, F points left — restoring force.

Right: v vs x is a semi-ellipse. Equation: $v = \pm\omega\sqrt{x_m^2 - x^2}$. Semi-axes are x_m (horizontal) and $v_m = \omega x_m$ (vertical).

At $x = 0$ (centre): $v = \pm v_m$ — fastest speed. At $x = \pm x_m$ (tips): $v = 0$ — block stopped at turning points.

The shape is an ellipse because rearranging gives $v^2/v_m^2 + x^2/x_m^2 = 1$, which is the standard ellipse equation.